May 4, 2026

Recall from last class: We want to build up:

Parametrized planes, parametrized surfaces, surface area, scalar surface integrals, vector surface integrals, orientation of a surface, reparametrization of a surface ...

in order to get to:

nice oriented surfaces as boundaries of nice regions in $\mathbb{R}^3$ and oriented closed simple curves in $\mathbb{R}^3$ as boundaries of nice surfaces to have: a generalization of the Fundamental Theorem of Calculus.

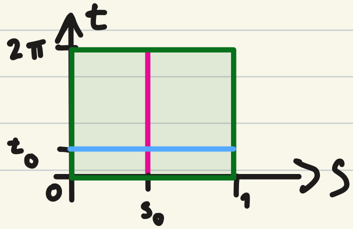
def: Let D be a region in $\mathbb{R}^2$ consisting of an <u>open connected</u> set (and possibly) together with some or all of its boundary points.

A <u>parametrized surface in $\mathbb{R}^3$</u> is a continuous function $\vec{X}: D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ which is one-to-one on all D, except possibly on ∂D (boundary).

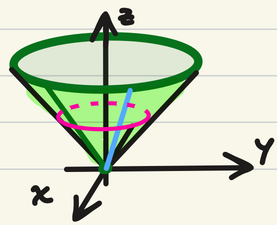
$S = \vec{X}(D) \subseteq \mathbb{R}^3$ is called a <u>surface</u> and $\vec{X}$ its <u>parametrization</u>.

def: For each constant s_0, t_0, the curves $\vec{X}(s_0, t)$ and $\vec{X}(s, t_0)$ are called coordinate curves.

e.g. $\vec{X}(s,t) = (s\cos t, s\sin t, s)$ $\quad 0 \leq s \leq 1, \ 0 \leq t \leq 2\pi$



— boundary



Tangent plane to a smooth parametrized surface

Remember we can describe a plane in $\mathbb{R}^3$ in two ways:
- Point and two tangent (direction) vectors, or
- Point and normal vector

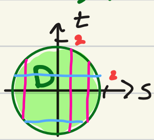
def: A parametrized surface $\vec{X}: D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ is said to be <u>smooth at $\vec{X}(s_0,t_0)$</u> if the map $\vec{X}$ is C^1 on an open ball at (s_0,t_0) and $\vec{N}(s_0,t_0) = \frac{\partial \vec{X}}{\partial s}(s_0,t_0) \times \frac{\partial \vec{X}}{\partial t}(s_0,t_0) \neq \vec{0}$.
(we call $\vec{N}(s_0,t_0)$ the <u>standard normal vector of the parametrization</u>)

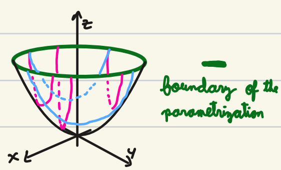
If a parametrized surface $\vec{X}: D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ is smooth at $\vec{X}(s_0,t_0)$, then the tangent plane to $S = \vec{X}(D)$ at $\vec{X}(s_0,t_0)$ is the plane that passes through the point $\vec{X}(s_0,t_0)$ and has normal vector $\vec{N}(s_0,t_0)$.

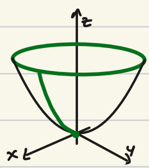
e.g. Last class we looked at two different parametrization of a subset of the paraboloid $z = x^2+y^2$

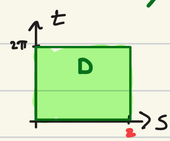
$\vec{X}(s,t) = (s, t, s^2+t^2)$ $\qquad$ $\vec{Y}(s,t) = (s\cos t, s\sin t, s^2)$

$D = \{(s,t) : s^2+t^2 \le 4\}$ $\qquad$ $D = \{(s,t) : 0 \le s \le 2,\ 0 \le t \le 2\pi\}$









$\vec{T}_s = \frac{\partial \vec{X}}{\partial s} = (1, 0, 2s)$ $\qquad$ $\vec{T}_s = (\cos t, \sin t, 2s)$

$\vec{T}_t = \frac{\partial \vec{X}}{\partial t} = (0, 1, 2t)$ $\qquad$ $\vec{T}_t = (-s\sin t, s\cos t, 0)$

$\vec{N} = \vec{T}_s \times \vec{T}_t = (-2s, -2t, 1)$ $\qquad$ $\vec{N} = (-2s^2\cos t, -2s^2\sin t, s)$

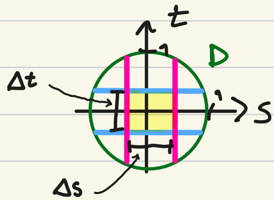
Finding tangent plane at $\vec{a} = (x,y,z) = (-\sqrt{2}, 0, 2)$

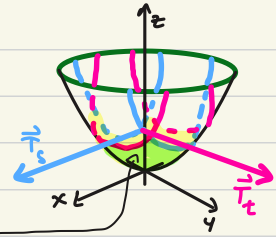
Three ways to find normal vector:
- Implicit surface: $0 = x^2+y^2-z$ $\quad$ gradient $(2x, 2y, -1)$ $\quad$ $\vec{N} = (-2\sqrt{2}, 0, -1)$
- Parametrization $\vec{X}$: $\vec{a} = \vec{X}(-\sqrt{2}, 0)$ $\quad$ $\vec{N} = (2\sqrt{2}, 0, 1)$
- Parametrization $\vec{Y}$: $\vec{a} = \vec{Y}(\sqrt{2}, \pi)$ $\quad$ $\vec{N} = (-2(\sqrt{2})^2\cos\pi, 0, \sqrt{2}) = \sqrt{2}(2\sqrt{2}, 0, 1)$

$\vec{Y}(s_0, t_0) = (-\sqrt{2}, 0, 2) \rightsquigarrow s_0^2 = 2 \rightsquigarrow s_0 = \sqrt{2}$ and $t_0 = \pi$ $\qquad$ $((x,y,z) - \vec{a}) \cdot \vec{N} = 0$

Surface area

geometric intuition: break down the domain D into rectangles, map the rectangles to the surface S and use the tangent vectors $\vec{T}_s, \vec{T}_t$ to approximate the area of that "patch".





Area of the "patch" $\approx \|\Delta s \vec{T}_s \times \Delta t \vec{T}_t\| = \|\vec{N}\| \Delta s \Delta t$

Surface area: If $\vec{X}: D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ is a smooth parametrization of the surface $S = \vec{X}(D)$, then the surface area of S is

$$\boxed{\iint_D \|\vec{N}(s,t)\| \, ds\,dt}$$

e.g. surface area of a sphere of radius R
Let's split the surface in two along the equator and compute the area for one hemisphere.

$$\vec{X}(r,\theta) = (r\cos\theta, r\sin\theta, \sqrt{R^2 - r^2}) \quad \text{north hemisphere}$$

$$D = \{(r,\theta) : 0 \le r \le R, \ 0 \le \theta \le 2\pi\}$$

$$\vec{T}_r = \left(\cos\theta, \sin\theta, \frac{-r}{\sqrt{R^2-r^2}}\right) \qquad \vec{T}_\theta = (-r\sin\theta, r\cos\theta, 0)$$

$$\vec{N} = \vec{T}_r \times \vec{T}_\theta = \left(\frac{r^2\cos\theta}{\sqrt{R^2-r^2}}, \frac{r^2\sin\theta}{\sqrt{R^2-r^2}}, r(\cos^2\theta + \sin^2\theta)\right)$$

$$\|\vec{N}\| = \sqrt{\frac{r^4}{R^2-r^2} + r^2} = \sqrt{\frac{r^2R^2}{R^2-r^2}} = \frac{r}{\sqrt{1 - r^2/R^2}}$$

$$\int_0^R \int_0^{2\pi} \frac{r}{\sqrt{1-r^2/R^2}} \, d\theta\, dr = 2\pi \int_0^R \frac{1}{\sqrt{1-r^2/R^2}} r\,dr \qquad u = 1 - \frac{r^2}{R^2}, \quad du = \frac{-2r}{R^2} dr \qquad = -\pi R^2 \int_0^R \frac{1}{\sqrt{1-r^2/R^2}} \left(\frac{-2r}{R^2}\right) dr$$

$$= -\pi R^2 \int_1^0 u^{-1/2} du = \pi R^2 \int_0^1 u^{-1/2} du = \pi R^2 \, 2u^{1/2} \Big|_{u=0}^{u=1} = 2\pi R^2$$

Surface area of the sphere: $4\pi R^2$ north hemi. + south hemi.

The graph of a function $f: D \subseteq \mathbb{R}^2 \to \mathbb{R}$ is the surface $z = f(x,y)$ in $\mathbb{R}^3$; we can compute its area:

$$\vec{X}(x,y) = (x, y, f(x,y))$$

$$\vec{T}_x = (1, 0, f_x), \quad \vec{T}_y = (0, 1, f_y), \quad \vec{N} = (-f_x, -f_y, 1)$$

$$\|\vec{N}(x,y)\| = \sqrt{(f_x)^2 + (f_y)^2 + 1}$$

surface area of the graph of f over D: $\boxed{\iint_D \sqrt{(f_x)^2 + (f_y)^2 + 1}\, dx\, dy}$

Scalar surface integral

def: Let $\vec{X}: D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ be a smooth parametrized surface, where D is a bounded region. Let $f: \mathrm{domain}(f) \subseteq \mathbb{R}^3 \to \mathbb{R}$ be a continuous scalar function with domain inside $\mathbb{R}^3$ containing the surface $S = \vec{X}(D)$, then the scalar surface integral of f along $\vec{X}$ is

$$\boxed{\iint_S f\, dS = \iint_D f(\vec{X}(s,t))\, \|\vec{N}(s,t)\|\, ds\, dt}$$

e.g. Integrate $f(x,y,z) = z^3$ over the parametrized sphere of radius R

$$\vec{X}(\theta,\varphi) = (R\cos\theta\sin\varphi,\ R\sin\theta\sin\varphi,\ R\cos\varphi) \qquad 0 \le \theta \le 2\pi,\ 0 \le \varphi \le \pi$$

$$\vec{T}_\theta = (-R\sin\theta\sin\varphi,\ R\cos\theta\sin\varphi,\ 0) \qquad \vec{T}_\varphi = (R\cos\theta\cos\varphi,\ R\sin\theta\cos\varphi,\ -R\sin\varphi)$$

$$\vec{N} = (-R^2\cos\theta\sin^2\varphi,\ -R^2\sin\theta\sin^2\varphi,\ -R^2\sin^2\theta\cos\varphi\sin\varphi - R^2\cos^2\theta\cos\varphi\sin\varphi)$$

$$\|\vec{N}\| = \sqrt{R^4\cos^2\theta\sin^4\varphi + R^4\sin^2\theta\sin^4\varphi + R^4\cos^2\varphi\sin^2\varphi} = R^2\sin\varphi$$

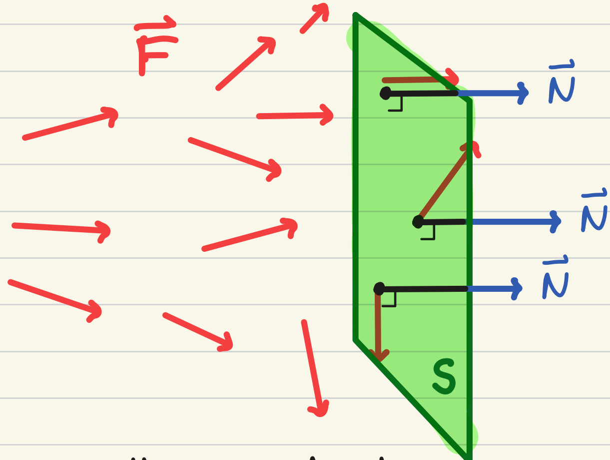
$$\iint_D f(\vec{X}(\theta,\varphi))\, \|\vec{N}(\theta,\varphi)\|\, d\theta\, d\varphi = \int_0^{2\pi}\int_0^{\pi} R^3\cos^3\varphi\ R^2\sin\varphi\, d\varphi\, d\theta$$

$$= 2\pi \int_0^{\pi} R^5\cos^3\varphi\sin\varphi\, d\varphi = 2\pi R^5 \left. \frac{-\cos^4\varphi}{4} \right|_{\varphi=0}^{\varphi=\pi}$$

$$= \frac{-\pi}{2} R^5 (1-1) = 0$$

Vector surface integral or flux

idea: Measure how much a vector field $\vec{F}$ is flowing across the surface.



def: Let $\vec{X}: D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ be a smooth parametrized surface and $\vec{F}: \text{domain}(\vec{F}) \subseteq \mathbb{R}^3 \to \mathbb{R}^3$ is a continuous vector field. Then, the flux of $\vec{F}$ across $S = \vec{x}(D)$ is the vector surface integral

$$\boxed{\iint_{\vec{X}} \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{X}(s,t)) \cdot \vec{N}(s,t)\, ds\, dt}$$

e.g. Compute the flux of $\vec{F}(x,y,z) = (4x, z^2, -z)$ across $\vec{X}(s,t) = (s-t, st, -s)$ $D = \{(s,t): 0 \le s \le 3, 0 \le t \le 1\}$.

$\vec{T}_s = (1, t, -1)$, $\vec{T}_t = (-1, s, 0)$

$\vec{N} = (s, 1, s+t)$

$\vec{F}(\vec{X}(s,t)) = (4(s-t), (-s)^2, -(-s))$

$$\iint_{\vec{X}} \vec{F} \cdot d\vec{S} = \int_0^3 \int_0^1 (4(s-t), s^2, s) \cdot (s, 1, s+t)\, dt\, ds$$
$$= \int_0^3 \int_0^1 6s^2 - 3st\, dt ds = \int_0^3 \left[6s^2 t - \tfrac{3}{2} s t^2 \Big|_{t=0}^{t=1} \right] ds$$
$$= \int_0^3 6s^2 - \tfrac{3}{2} s\, ds = 2s^3 - \tfrac{3}{4} s^2 \Big|_{s=0}^{s=3} = \left(2 - \tfrac{1}{4}\right) 3^3 = \frac{7 \cdot 27}{4}$$